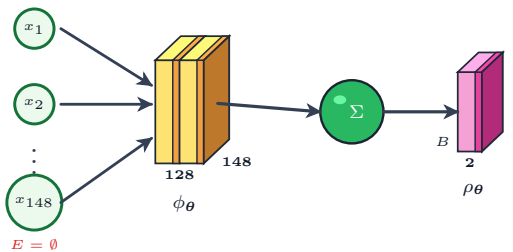
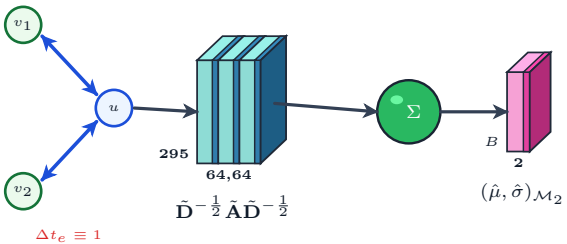
1. M1_MLP_DeepSets (Zaheer et al., 2017): $E = \emptyset$



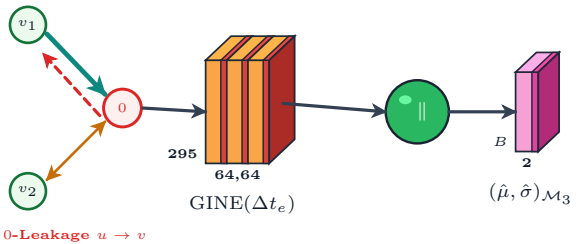
$(\hat{\mu}, \hat{\sigma})_{\mathcal{M}_1} = \rho_{\boldsymbol{\theta}}\left(\bigoplus_{i=1}^{148} \phi_{\boldsymbol{\theta}}(x_{k,i}, T_i)\right) \implies R^2(\hat{\sigma}) =$ **0.5686**, $\mathrm{MSE}(\hat{\mu}) = 0.6844$

2. M2_GCN_Unweighted (Kipf & Welling, 2017)



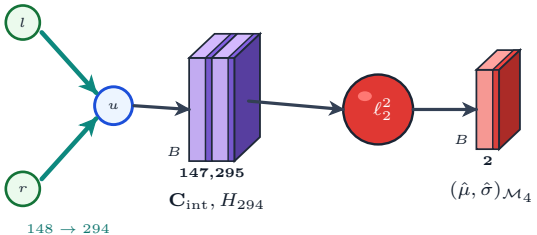
$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \implies R^2(\hat{\sigma}) =$ **0.8867**, $\mathrm{MSE}(\hat{\mu}) = 0.5859$

3. M3_EdgeMPNN_GINE (Hu et al., 2020): $u \leftrightarrow v$



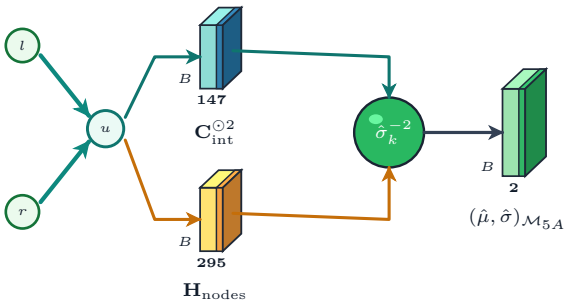
$\mathbf{h}_v^{(l+1)} = \psi\left((1+\epsilon)\mathbf{h}_v^{(l)} + \sum_{u \in \mathcal{N}(v)} \mathrm{ReLU}(\mathbf{h}_u^{(l)} + \phi_e(\mathbf{e}_{uv}))\right) \implies$
$R^2(\hat{\sigma}) =$ **0.9790**

4. M4_PhyloGNN_RawMSE: Directed $v \to u$ + $\mathcal{L}_{\mathrm{MSE}}$



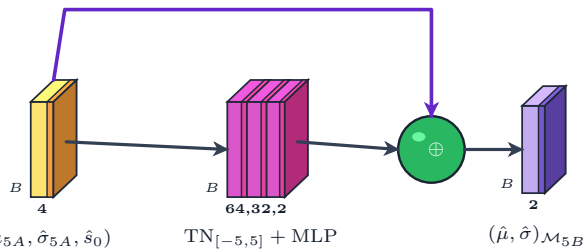
$\mathcal{L}_{\mathrm{MSE}} = \|\hat{\mu} - \mu\|_2^2 + \|\hat{\sigma} - \sigma\|_2^2 \implies$
2,500× Grad Bias, $\mathrm{MSE}(\hat{\mu}) =$ **0.7933**

5. M5A_PhyloGNN_UnbiasedNLL: Dual-Stream + $\hat{\sigma}_k^{-2}$



$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{2}\left[\frac{(\hat{\mu}_k - \mu_k)^2}{\hat{\sigma}_k^2} + \ln \hat{\sigma}_k^2\right] \implies R^2(\hat{\sigma}) =$ **0.9966**, $\mathrm{MSE}(\hat{\sigma}) =$ 0.01488 (phylolm)

6. M5B_PhyloGNN_BayesMMSE: Empirical Bayes



$\hat{\mu}_{5B} = \mathbb{E}[\mu \mid \mathbf{x}_k]_{[-5,5]} \implies$
−30.2% $\mathrm{MSE}(\hat{\mu})$ (**0.5311**), **−13.1%** $\mathrm{MSE}(\hat{\sigma})$ (**0.01293**)